Javascript

Arrays,functions

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Functions</title>
  </head>
  <body>
    <h1>Functions</h1>

    <script>
       let a =10;
       let b = 10;

       function add(a, b){ // parameter
            console.log("add function ");
            console.log(a+b);
            return a+b;
        }
        add(30, 40); // argument
       console.log(a+b);
       let c = 10;
       let d = 10;

       let e=10;
       let r=10;

       let res = add(e, r);
       console.log(res);

    //     ES6  arrow function
     let sub =(a, b)=>{
        console.log("Sub function" , a-b);
     };
     sub(10, 40);

    //  Anonymouse function   IIFE (immediate invoked function)

    (function iife(){
        console.log("Immediate invoked function");

    })()

    // function expression
```

```
let express = function exp(){
    console.log("Function expression");
}
express();


function user(name="Academy", age=10){
    console.log(name, age);
}
user();


function rest(...restparameter){
    console.log(restparameter.includes("admin"));

}
rest("admin", "user");
// int arr[] =new int[5]; in java

let arr =[1,2,3,4,2,5]; // continguous memory, also accept non-similar
types
console.log(arr);

for(let i=0; i<arr.length; i++){
    console.log(arr[i])
}
console.log(arr[0]);
console.log(arr[1]);
console.log(arr[3]);
console.log(arr[4]);
// console.log(arr[5]);
// console.log(arr[6][1]);
arr[2] = 20;

console.log(arr[2]);

// array methods => push, pop, shift, unshift, slice

console.log(arr.push(6));
console.log(arr.push(7));
console.log(arr.pop());

console.log(arr.unshift(10));
console.log(arr.shift());


console.log(arr.slice(2,5));  // n-1 -> 3-1 = 2
```

```
    console.log(arr);

    console.log(arr.length);

    console.log(arr.lastIndexOf(2));
    console.log(arr.indexOf(2));

    console.log(arr.splice(3,4));
    console.log(arr);

    console.log(arr.includes(2));

    </script>
  </body>
</html>
```

Theory Notes

1. Functions

- **Definition**: A block of reusable code that performs a specific task.
- **Types**:
  - **Named functions** → declared with function keyword.

function add(a, b) { return a+b; }

  - **Arrow functions (ES6)** → shorter syntax, often used for callbacks.

let sub = (a, b) => a-b;

  - **Anonymous functions** → functions without a name, often assigned to variables.

let exp = function() { console.log("Expression"); };

  - **IIFE (Immediately Invoked Function Expression)** → runs immediately after definition.

(function(){ console.log("IIFE"); })();

- **Parameters vs Arguments**:
  - Parameters → placeholders in function definition.
  - Arguments → actual values passed when calling the function.

2. Default Parameters

- Functions can have default values for parameters.

```
function user(name="Academy", age=10) { console.log(name, age); }
```

3. Rest Parameters

- Collects multiple arguments into an array.

```
function rest(...values) { console.log(values); }

rest("admin", "user");
```

4. Arrays

- **Definition**: Ordered collection of values.
- **Properties**:
  - length → number of elements.
  - Indexed starting at 0.
- **Methods**:
  - push() → add element at end.
  - pop() → remove last element.
  - shift() → remove first element.
  - unshift() → add element at start.
  - slice(start, end) → extract portion (non-destructive).
  - splice(start, count) → remove/replace elements (destructive).
  - indexOf() / lastIndexOf() → find element position.
  - includes() → check if element exists.

Practice Tasks

Short Answer

1. What is the difference between parameters and arguments?
2. Why are arrow functions often used in callbacks?
3. What is the difference between slice() and splice()?
4. How does includes() differ from indexOf()?

Coding Exercises

1. Write a function to calculate the square of a number.
2. Create an arrow function to subtract two numbers.
3. Implement an IIFE that prints "Hello Academy".
4. Use rest parameters to accept multiple names and print them.
5. Create an array of fruits and demonstrate push, pop, shift, and unshift.
6. Use slice() to extract a portion of an array.
7. Use splice() to remove and replace elements in an array.